

Module 2: Intelligent Data Driven Applications on the Cloud – FINAL PROJECT

INTELLIGENT DATA DEVELOPMENT, JUNE 2026

FINAL PROJECT - DEPLOYING A FORECASTING MODEL FOR GLOVO

- **Team:** Groups of 2 students. Groups of 3 allowed only by exception.
- **Due date:** 28th of June
- **Evaluation:** The final project represents 60% of the module grade. The four project deliverables are equally weighted within the final project.
- **Project repository:** <https://github.com/javiermas/bse-ts-public>
- **AI policy:** AI allowed with disclosure, validation, and shared prompt history.

Project: The goal of the project is to deploy an AWS cloud solution that trains a model and generates predictions of orders using Glovo data.

Deliverables:

- **Code**
 - In any format, as long as it runs in the cloud. You may use any cloud-based compute environment of your choice.
- **Video (max 10 minutes)**
 - A walkthrough of your code and cloud setup. Ideally, you run the code on the cloud and generate an output file in the video. If your job takes longer than 10 minutes, you don't need to show the entire process—use caching where helpful. For recording the video, you may use any tool if you can share a link or file. If your tool limits video length, you can submit two clips.
- **Predictions** – A .csv file with out-of-sample predictions (format described below), stored in S3
- **Prod deployment considerations**
 - Section (PDF – max 5 pages) describing concepts thought in this part of the course: Define team(s) and their roles, interactions and ways of working. Define rough estimates for each part/ feature of the project and how these estimates were agreed. Define the architecture components of the end-to-end solution. Define data used and data flows (i.e. how data travels from the Storage/Database to your “forecasting tool” and connects to a visualization tool). Describe and argument environments to be used. Describe how the build and monitor of the ML Model will happen. Describe how the end-users will use your solution (visualization tool, interactive platform, etc.). Describe success KPIs.

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Business requirements

You've just joined Glovo. The company uses a **slot-based system**, dividing each day into **24 hourly slots** per city. For efficient operations, Glovo needs to predict the optimal number of couriers required in each slot. Overestimating leads to idle couriers and wasted resources; underestimating causes delivery delays.

Currently, Operations manually determines these numbers based on past demand, but as Glovo expands, this manual process no longer scales. Your task is to help automate it by **forecasting hourly order volumes**.

To simplify, we'll focus on **one city only**. Every **Sunday at 23:59**, your model should forecast the **number of orders expected in each hourly slot** for the upcoming week (from **Monday 00:00 to Sunday 23:00**, inclusive). You may use all data up to and including the current week to generate your forecast.

EDA

- The data is under *data/train_data.csv*. Explore the data, visualize it. Look for trends, cycles and seasonality. Also, can you find any outliers? How does your EDA inform your modelling?

Modelling

- Try different models. Validate each model in a way that would imitate the real problem (every Sunday you forecast all next week). Watch out for data leakage. Evaluate each model on MSE and SMAPE. Which one performs better?

Output format

- A .csv file with columns: time (hourly resolution), datetime64[ns] type and preds: float64 type, 24*7 rows (7 days worth of predictions)

Going from **2022-01-24 00:00:00** to **2022-01-30 23:00:00** (both inclusive)

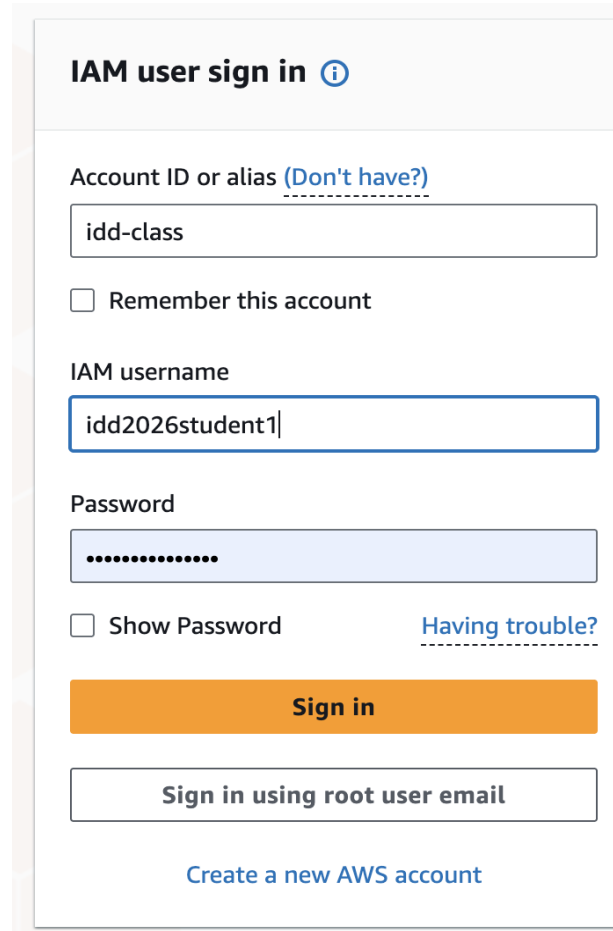
- You can check your output format with the function provided to you in the github repository. There might be some check we've missed, so make sure the output is what you want it to be.

CONNECT TO THE AWS ACCOUNT USING AN IAM USER

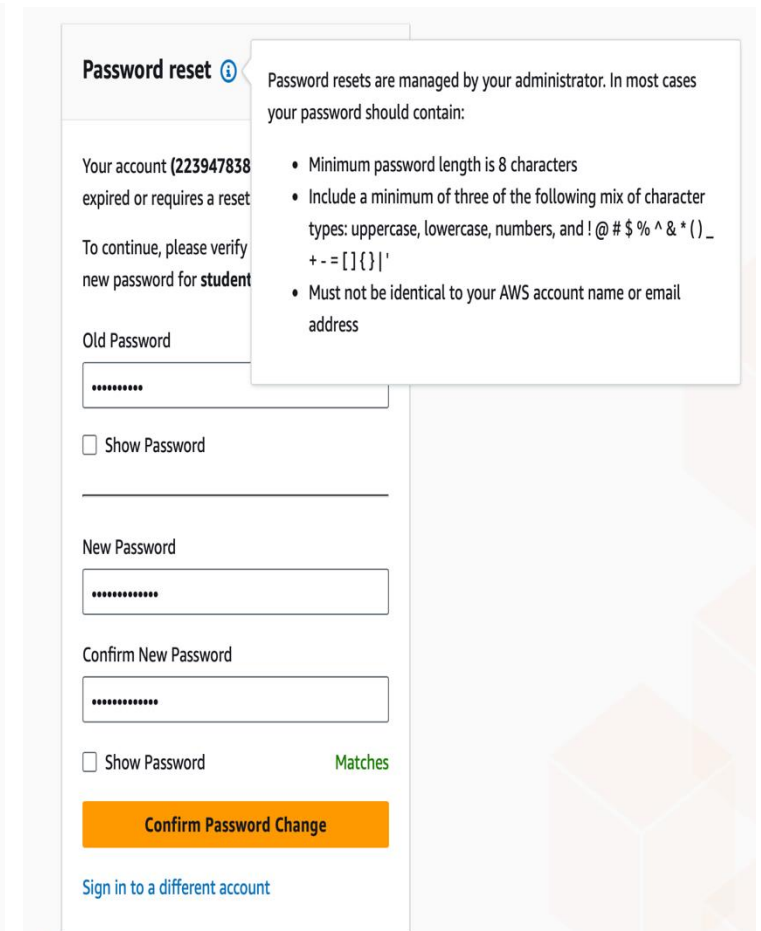
Step-by-step:

1. Go to the login link: **<https://idd-class.signin.aws.amazon.com/console>**
2. Log in using the credentials:
 - **Username:** *name.surname/*
name.middlename (or your assigned name on the next slide)
 - **Password:** *ChangeMeNow_2024!*
3. On first login, set a **new password** you'll remember. A password reset screen will appear. You will need to add the old password provided and add your new one.

You're now inside your AWS account!



The screenshot shows the 'IAM user sign in' page. It has a title 'IAM user sign in' with an information icon. Below the title, there's a section for 'Account ID or alias' with a link '(Don't have?)' and a text input field containing 'idd-class'. There's a checkbox for 'Remember this account'. Below that is the 'IAM username' section with a text input field containing 'idd2026student1'. The 'Password' section has a text input field with masked characters. There's a checkbox for 'Show Password' and a link 'Having trouble?'. At the bottom, there are two buttons: 'Sign in' (orange) and 'Sign in using root user email' (white with orange border). At the very bottom, there's a link 'Create a new AWS account'.



The screenshot shows the 'Password reset' page. It has a title 'Password reset' with an information icon. Below the title, there's a message: 'Your account (223947838) expired or requires a reset. To continue, please verify new password for student'. There's a list of password requirements: 'Minimum password length is 8 characters', 'Include a minimum of three of the following mix of character types: uppercase, lowercase, numbers, and ! @ # \$ % ^ & * () _ + - = [] { } | ' ', and 'Must not be identical to your AWS account name or email address'. Below this is the 'Old Password' section with a text input field containing masked characters and a checkbox for 'Show Password'. The 'New Password' section has a text input field containing masked characters. The 'Confirm New Password' section has a text input field containing masked characters and a checkbox for 'Show Password'. There's a green 'Matches' indicator. At the bottom, there are two buttons: 'Confirm Password Change' (orange) and 'Sign in to a different account' (blue link).

USERS

Student

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galla-gelpi-buxade
gavin-robert-budd
grace-alden-hotchkiss
jillian-ann-hunter
joshua-ramas-castillo
luis-david-batista-tome
maria-victoria-suriel-nunez
mariajose-argote-corrales
matthieu-david-ludovic-hildreth
nicolas-aris-koutalakis
olsa-berani
olta-recica
ronnie-kwadwo-obeng-osei
samuel-gerard-fraley
sebastian-dong-uk-paik-sohn
sofia-pirogova
tizian-schenk
xiaoyan-wang

USEFUL INFORMATION

1. **S3 buckets:** <name>-<surname>-bucket-idd, i.e. laura-cozma-bucket-idd
 1. Please upload ***test_data_mock.csv*** and ***train_data.csv*** before trying to test the lambda function.
2. **Lambda function:**
 1. Name: <name>-<surname>-forecasting-lambda
 2. Runtime: **Python 3.X**
 3. Execution role: assigned
 4. Layers: pandas, skitlearn